EXPERIMENT 4

Document Storage and Top-k Retrieval using a Vector Database

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AIM

To store document vectors in a FAISS vector database and retrieve the top-k closest documents for a query.

PROCEDURE

1. Import FAISS and NumPy.
2. Prepare sample documents and illustrative vector embeddings.
3. Create a FAISS index and store the document vectors.
4. Set a query vector and the required top-k value.
5. Search the FAISS index.
6. Display the top-k retrieved documents with their distances.

PROGRAM

```
# Install first: pip install faiss-cpu numpy
import faiss
import numpy as np

documents = [
    "Python is easy to learn.",
    "AI is used in smart machines.",
    "SQL manages databases.",
    "Machine learning learns from data."
]

# Illustrative document embeddings
vectors = np.array([
    [1.0, 0.0, 0.0],
    [0.0, 1.0, 0.0],
    [0.0, 0.0, 1.0],
    [0.0, 0.9, 0.1]
], dtype="float32")

index = faiss.IndexFlatL2(3)
index.add(vectors)

query_vector = np.array([[0.0, 1.0, 0.0]], dtype="float32")
k = 3

distances, indices = index.search(query_vector, k)

print("Top", k, "Retrieved Documents:")
for rank, doc_id in enumerate(indices[0], start=1):
    print(rank, documents[doc_id],
          "Distance:", round(float(distances[0][rank - 1]), 3))
```

OUTPUT

```
# Search for top 3 documents
query = "How does AI learn from data?"
query_vector = vectorizer.transform([query]).toarray()
query_vector = np.array(query_vector, dtype="float32")

distances, results = database.search(query_vector, 3)

print("Query:", query)
print("\nTop 3 Retrieved Documents:")

for rank, idx in enumerate(results[0], start=1):
    print(rank, documents[idx])
```

```
Requirement already satisfied: faiss-cpu in /usr/local/lib/python3.13/dist-packages (1.15
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.13/dist-packages (1
Requirement already satisfied: numpy>=1.25 in /usr/local/lib/python3.13/dist-packages (fr
Requirement already satisfied: packaging in /usr/local/lib/python3.13/dist-packages (from
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.13/dist-packages (f
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.13/dist-packages (
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.13/dist-pac
Requirement already satisfied: cloudpickle>=3.0 in /usr/local/lib/python3.13/dist-package
Query: How does AI learn from data?

Top 3 Retrieved Documents:
1 Machine learning learns from data
2 AI helps machines make decisions
3 Python is used for programming
```

RESULT

Thus, documents were stored in a FAISS vector index and the top-k nearest documents were retrieved successfully.